

Reference architecture design for computer-based speech therapy systems

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ABSTRACT

With the current international shortage of speech-language pathologists (SLPs), there is a demand for online tools to support SLPs with their daily tasks. For this purpose, several online speech therapy systems (OSTSs) have been proposed and discussed in the literature. However, developing these OSTSs is not trivial since it involves the consideration of various functional and quality concerns. Hence, for communicating the design decisions and guiding the development and analysis of these systems, a proper architecture design is important. Unfortunately, the architecture design of OSTSs has not been explicitly addressed in the literature. To this end, we present a reference architecture for OSTSs which has been designed following well-established architecture design methods. The reference architecture captures the reusable design elements of OSTSs and can be used to derive various different application architectures. A case study approach is used to illustrate and validate the use of the presented reference architecture.

1. Introduction

Speech therapy has been a widely discussed practice and topic worldwide and a recommended treatment for various forms of speech disorders (Palmer and Enderby, 2007). A speech disorder is defined as a problem with fluency, voice, and how a person pronounces speech sounds and is often referred to as a phonological disorder (Danubianu et al., 2009). Speech disorders are communication disorders that disrupt everyday speech (Law et al., 2017). According to the American Speech-Language-Hearing Association (ASHA)¹ a communication disorder is "an impairment in the ability to receive, send, process, and comprehend concepts or verbal, nonverbal and graphic symbol systems. A communication disorder may be evident in the processes of hearing, language, and/or speech". The ASHA states that speech therapy can be due to an impairment of the articulation of speech sounds, fluency, language and/or voice (ASHA, 2021). One of the first speech therapy books was written almost 100 years ago by Elizabeth Brehm in 1922 entitled "speech correction" (Brehm, 1922). Surprisingly, not much seem to have changed in this practice since then (Towey, 2012). Nevertheless, 21st-century speech practitioners have started to rely more on computer applications during therapy. Although digitised, many of these applications are still based on the conventional practice types where the services provided are still generic and not personalised (Towey, 2012). This is unfortunate as new technologies provide opportunities to create more personalised treatment plans. Children in middle-to-high income economies are already exposed to technology at an early age and, therefore, quickly comprehend and embrace the online process of using software systems (Ben-Aharon, 2019). More than 40 percent of the clients of speech therapists are children with speech sound disorders (Furlong et al., 2017). However, not all children that need speech therapy currently have access to a therapist. Especially in lower-income economies, there is a lack of therapists or the means

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¹ <https://www.asha.org/practice-portal/clinical-topics/articulation-and-phonology/>

Nomenclature

Acronyms

AA	Application Architecture
ADC	Analogue-to-digital converter
ASR	Automatic Speech Recognition
CBST	Computer-based speech therapy
MIS	Management information system
ML	Machine Learning
OSTS	Online Speech Therapy System
RA	Reference Architecture
SLP	Speech-language pathologist
UI	User interface

to afford one (Franciscatto et al., 2021). Such deprivation of needed guidance can damage a person's professional ambitions and lower their social contacts (Franciscatto et al., 2021). The usual routine of working with several clients a day makes it challenging for the speech therapist to provide an individually tailored therapy plan at the end of the session (Robles-Bykbaev et al., 2015). The task of a speech therapist is broad and does not only entail doing exercises with the patient, as it is often believed. As the ASHA states: "Speech-language pathologists work to prevent, assess, diagnose, and treat speech, language, social communication, cognitive-communication, and swallowing disorders in children and adults" (ASHA, 2021). The conventional method of meeting a therapist physically thus needs to be improved or even automated utilising an Artificial Intelligence-based solution.

Computer-based speech therapy systems could help to resolve service access problems, particularly for families having limited access to therapy (Furlong et al., 2017). Previous research acknowledges the positive impact that the usage of Information and Communication Technologies has on both the therapist and the patient during therapy (Toki et al., 2012). Recent scientific literature shows a variety of currently existing online speech therapy systems (OSTSs). Some recent examples are "Into the Forest" (Nasiri and Shirmohammadi, 2017), "AACVOX" (da Silva et al., 2018), "Tingog" (Belen et al., 2018) "Apraxia World" (Hair et al., 2018), "AppVox" (Rocha et al., 2019), "Talking Kids" (Firdous et al., 2019), "WordWise", and others.

Similar to other software systems, developing these OSTSs requires the proper Internet of Things architecture which represents the main structure of the system. An architecture comprises software elements and how the elements relate to each other (Bass et al., 2003). Software architecture breaks the software design into structures based on the software requirements. OSTSs typically have different functional requirements such as the progress evaluation, feedback and reward, and the data processing capacity. Besides functional requirements, quality requirements such as security level, safety level, time performance, and overall cost of development and operation are different for different applications. The successful development of these computer-based speech therapy programs which are domain specific is very much dependent on the right reference architecture (RA). The RA is derived from the requirements analysis documents and this is a very daunting task to derive. A reference architecture is a generic design that can derive specific application architectures (AAs) based on the identified stakeholder concerns (Tekinerdogan et al., 2005; Kassahun et al., 2016; Tummers et al., 2021). To the best of our knowledge, no study exists on how OSTSs are designed and has developed a reference architecture for the OSTS domain according to the software architecture documentation guidelines. Therefore, this study focuses on designing a RA for computer-based speech therapy programs that can support a speech therapist and can be operated from a personal computer (Furlong et al., 2017). This study aims to develop a validated RA dedicated to OSTSs following well-established architecture design methods. The main research question of this study is defined as follows:

- RQ: What is a proper reference architecture for OSTSs?

To design the reference architecture, a domain-driven architecture design approach (Tekinerdogan and Aksit, 2002; Evans, 2014) is adopted as presented in the software architecture literature (Martin, 2000). By conducting a thorough domain analysis, we provide an architecture model for a broad domain of OSTSs. Additionally, we conducted a case study at a professional speech and language therapy company based in the Netherlands to evaluate the proposed RA.

The rest of this article is organised as follows: Section 2 provides some brief background on OSTSs. Section 3 explains the methodology and RA features and Section 4 presents the derived reference architecture. Section 5 presents the results of the case study research. Section 6 covers the discussion. Finally, Section 7 presents the conclusions.

2. Background and related work

This section describes the background on OSTSs and related works concerning the use of architectures for OSTSs, general RA design and documentation.

2.1. Online speech therapy systems

A wide variety of OSTs have been proposed to support speech therapy activities (Chen et al., 2016; Darejeh et al., 2019; Furlong et al., 2017). The adoption and use of computers for speech therapy are called computer-based speech therapy (CBST) (Furlong et al., 2017). The following is a definition of CBST given by Furlong et al. (2017) (p.51): “A CBST program is a software offering predefined therapy tasks inclusive of instructional features, motivational features and quantitative features, operating from a personal computer”. For this study, we consider an OSTs as a management information system (MIS) used for collecting, processing, storing, and disseminating data in the form of information needed to carry out the operational functions typically done in a speech therapy clinic. Watson and colleagues (1991) defined a MIS as “an organisational method of providing past, present and projected information related to internal operations and external intelligence”.

2.2. Architecture design

A software system is composed of several structures and every software system needs an architecture that portrays its structure. Bass et al. (2003) define software architecture as a set of structures needed to reason about the system, which comprise software elements, relations among them, and properties of both. In the book of Bass et al. (2003), they note that a structure is architectural if it supports reasoning about the system and the qualities of the system. Software architecture serves as a bridge between software requirements and source code. Software architectures such as product line architectures and RAs have been proposed for different purposes (Nakagawa et al., 2011; Angelov et al., 2012). The U.S. department of defense defines a RA as “an authoritative source of information about a specific subject area that guides and constrains the instantiations of multiple architectures and solutions (US Dept of Defence, Office of the DoD CIO, 2010). A RA can be used as a reference for developing an AA. An AA is a software model of a particular application made by combining architectural views (Demirli and Tekinerdogan, 2011). This type of architecture is also called a concrete architecture (Angelov et al., 2012). So, in other words, a RA can be used to guide the development of a concrete architecture and thus play a prescriptive role, while the AA is descriptive for the RA. Therefore, a RA is made first, which is then used to develop an AA (Tummers et al., 2021). An AA represents the architecture of a single system (Tekinerdogan and Uzun, 2019), in this case, an OSTs.

2.3. Related works

Few prior studies discuss the development of OSTs with different system architectures. Most of these studies (Cagatay et al., 2012; Danubianu et al., 2010; Jamis et al., 2019; Origlia et al., 2018; Parmanto et al., 2008; Parnandi et al., 2013; Robles-Bykbaev et al., 2015) design and develop expert systems to generate therapy plan for speech and language disorders. Our review discovered three studies that provided extensive explanation of their displayed architecture. Ochoa-Guaraca et al. (2016) and Redrovan-Reyes et al. (2019) proposed a general system architecture for their systems. Franciscatto et al. (2021) presented a more detailed software architecture. Detailed information on these studies are discussed in this section.

(Franciscatto et al., 2021) proposed a software architecture to predict phonological processes for children speaking Portuguese to support speech therapists in their clinical decisions. They present two different modules, namely a Capture Module and a Service Module. The Capture Module collects the speech data from the client with the help of audio recording. Subsequently, the speech data is sent to the Service Module, consisting of a Classification Module and a Case-Based Reasoning (CBR) Cycle Module. The Classification Module identifies the received speech as correct or incorrect through pre-processing, spectrogram generation (audio to image) and feature extraction. After classification, the CBR Cycle Module tries to understand the error patterns in the pronunciation based on the retrieval of existing cases from the system's phonological knowledge base. For each module discussed, they presented a layout showing how components were related to each other. When a system prototype was evaluated with the help of speech samples collected from 1114 evaluations for 84 Portuguese words, the average accuracy of classifying pronunciations was 92.5 percent (Franciscatto et al., 2021). Ochoa-Guaraca et al. (2016) described a general system architecture for a robotic assistant to support different activities conducted by speech therapists during therapy sessions for Spanish-speaking children. They presented three layers needed for their system: expert system, user interface and service and knowledge domain management. Furthermore, the main modules for each layer were provided. The UI and services layer provided several functionalities for helping the speech therapists. The expert system layer relied on two modules for performing the processes to interact with users and support the decision-making of speech therapists. The domain knowledge was managed in the last layer through ontologies, databases for monitoring and activities, standardised vocabularies, and a clinical data repository (Ochoa-Guaraca et al., 2016). They also described the robot's electronic design and the main components of the hardware, such as a touchscreen. However, the model in their study did not follow formal modelling techniques. Similar to the work of Ochoa-Guaraca et al. (2016), Redrovan-Reyes et al. (2019) provided a general system architecture with three main layers: interaction layer, services layer and knowledge layer. The interaction layer of the system in the study has two apps, one that the therapist can use to manage the client's exercises and one to perform administrative tasks. The information captured in the interaction layer is sent to the expert system in the service layer, which contains the following modules: the web services, a user management module, a report generation module, and a module for speech recognition.

Table 1
Dimension values of the RA in this study.

	Dimension	Values
Context	C1: Where will it be used?	Single organisation
	C2: Who defines it?	Software design groups, User groups
	C3: When is it defined?	Classical
Goals	G1: Why is it defined?	Facilitation
Design	D1: What is described?	Components, policies/guidelines
	D2: How detailed is it described?	Semi-detailed components, Semi-detailed guidelines
	D3: How concrete is it described?	Semi-concrete elements
	D4: How is it represented?	Semi-formal element specifications

2.3.1. Conclusion of current reference architectures

According to the classification of [Angelov et al. \(2012\)](#), the previously discussed architectures in the related works section are not reference architectures and can be considered as informal and fragmented architectures as all the studies use ill-defined graphical notations or natural languages. Notably, we could not find a study in which a RA was discussed in a semi-formal (well-defined notation such as UML) or even formal wise (based on an architecture specification language such as C2 or Rapide). [Garlan and Schmerl \(2007\)](#) discussed some of the benefits of following a formal architecture model in their research. They noted that adopting and using a formal architecture model ensures the development team and requirements are organised at a high level before coding thus making it easier for future team members to easily maintain the software application. Additionally, [Bass et al. \(2003\)](#), [Qin et al. \(2008\)](#) highlights the benefits of having a formally designed architecture for software development which includes significantly savings on time and cost. However, even though we aim to develop a semi-formal RA, the previously discussed non-formal architectures are still helpful. We learn from the practical descriptions and use the information for building a concrete RA that can ideally be reused. We thus design and develop a formal RA which is generic, systematic and aligned with software architecture documentation guidelines. Since no prior study did design and develop a RA, it is impossible to compare and evaluate our RA with others in the same speech therapy system domain.

3. Research methodology

In this section, we follow well-documented approaches and guidelines adopted from the software engineering literature ([Evans, 2014](#)) for the domain analysis and designing of the RA. In the book of [Evans \(2014\)](#), they provide several useful guidelines on how to systematically design software for very complex domains. They provide a reference for domain-driven design that focuses on the knowledge of the team in a specific domain thus ensuring very delicate and intricate problems are easily identified and modelled for a robust software design. Using this well-documented approach will help us conduct a feasible and efficient domain analysis of an online speech therapy software system. A domain analysis is necessary for the development of any major software system. The objective is to precisely analyse the requirements of the project, compare and equate these requirements to already available prototypes that can be reused. This is conducted to find and mitigate high-risk areas of the project and describe poorly defined requirements ([Nielsen, 2014](#); [Tekinerdogan et al., 2019](#)). We present the domain analysis of a typical online speech therapy system with the help of a three-layer architecture and feature model. Researchers involved in this study are experienced software engineers and architects who have been applying these approaches for over a decade.

3.1. Context, goals and design dimensions

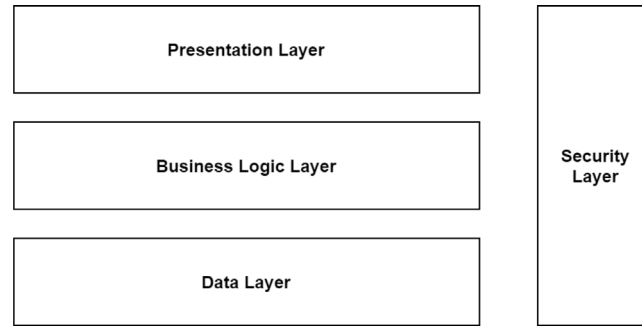
Usually, an architecture is used to develop a specific software application in a particular domain ([Bass et al., 2003](#); [Kang et al., 1998](#); [Angelov et al., 2009](#)). A domain can be described as a set of “common” problems that applications in that domain can do ([Angelov et al., 2012](#); [Nakagawa et al., 2011](#)). Each RA has its own context, goal, and design. In the next section, we describe the RA’s context, goal, and design dimensions in an OSTs domain, based on the work of [Angelov et al. \(2012\)](#). The values of each dimension are presented in [Table 1](#). By adopting and following the guidelines of [Angelov et al. \(2012\)](#) and [Evans \(2014\)](#), the RA we develop in this research is classified as type 4.

The RA designed in this study is initially aimed at a single organisation. Thus, the stakeholders are the organisational groups that are in a way related to this company. The single company is the recipient of the RA. Based on the prior works discussed in this study, we identified seven stakeholders and their concerns, represented in [Table 2](#). Research on OSTs has attracted several interests throughout the past years leading to the proposal and development of several OSTs. We aim to design a classical RA where we use parts of publicly available commercial systems ([Angelov et al., 2012](#)), thus saving much time, as we do not have to invent the technology, software, and algorithms required. The goal of this RA is to facilitate the design of a concrete architecture by providing means of inspiration and specific requirements for the creation of the system. A concrete architecture is based on the needs of the stakeholder concerns and the related requirements. We developed a semi-concrete architecture that specifies a specific class of options for each element in the architecture ([Angelov et al., 2012](#)).

Table 2

The identified stakeholders and their concerns.

Role	Concerns
Therapy Clinic Manager	Have a reliable working system that makes it possible to provide speech therapy to more clients who otherwise must wait.
Speech therapist	Save time on administrative tasks by using an OSTs.
OSTS developer	Develops the OSTs and its underlying software with the help of the developed blueprint.
Administrator	Can set up the system and manage the OSTs. Is not necessarily the OSTs developer.
Child	Enjoy the online therapy sessions while improving speech.
Parent	Receive safe and suitable working online therapy sessions for their child.
Researchers	Develop a blueprint for the OSTs.

**Fig. 1.** OSTs layered architecture.

3.2. Stakeholders and stakeholders' concerns

The definition of a stakeholder given by the project management institute was adopted. According to the project management institute, a stakeholder is “an individual, group, or organisation, who may affect, be affected by, or perceive itself to be affected by a decision, activity or outcome of a project” (Institute, 2013). The interests of stakeholders are expressed as concerns as presented in Table 2.

3.3. Layered architecture

Various studies (Cagatay et al., 2012; Danubianu et al., 2010; Franciscatto et al., 2021; Jamis et al., 2019; Ochoa-Guaraca et al., 2016; Origlia et al., 2018; Parmanto et al., 2008; Parnandi et al., 2013; Redrovan-Reyes et al., 2019; Robles-Bykbaev et al., 2015) have provided architectures for OSTs. Studies that elaborate a more extensive architecture seem to use a client–server approach, sometimes with layers. Based on the literature (Franciscatto et al., 2021; Ochoa-Guaraca et al., 2016; Redrovan-Reyes et al., 2019), we provide a layered RA shown in Fig. 1.

3.4. OSTs family feature model

Fig. 2 provides a family feature model for an OSTs, called Talking Heroes, that contains typical features of OSTs domains. The features have been derived from a domain analysis process in which we looked at the key literature in which common features were identified. The definition of these features is based on the explanation given in these prior studies. A reuse-based approach was used in which common and variant features have been represented. The top-level feature diagram corresponds with the OSTs Layered Architecture that is previously portrayed in Fig. 1. The presentation layer contains the Interface and Visualisation feature, which comprises the interfaces and sub-features for the client and speech therapist. The business logic layer is responsible for data processing and data analysis. These features are discussed below.

3.4.1. Client interface

The display presented to clients when the OSTs is accessed. This interface includes the following modules.

- Feedback: During the exercises, the clients must receive feedback to learn from the mistakes they made and also to commend them. This feedback can be in the form of an audio, visual, or textual. Below, we discuss the different feedback formats.
 - Audio Feedback: Audio feedback can be in the form of reading out hints, instructions on how to play, or saying the correct answer if the child gave the wrong answer. The system compares the user voice with the correct pronunciation and replies with a sound message: “Congratulations, that was the correct answer”, or “Wrong word, please try again” if the answer was not good enough.

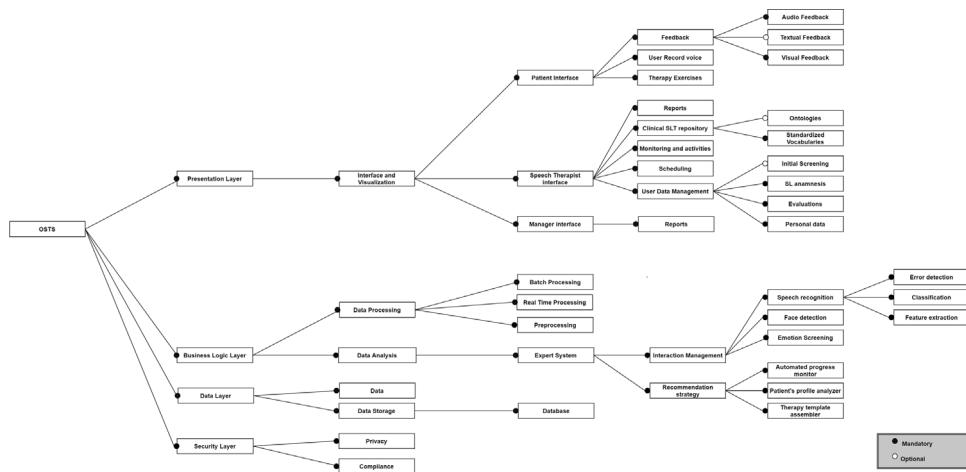


Fig. 2. OSTS feature diagram.

- Textual Feedback: What is said by the system can also be textually shown on the display of the computer when necessary, for hearing impaired clients. However, there should be an option to turn this off as younger children might not be literal yet, and for them, the sudden appearance of text might be confusing.
- Visual Feedback: Refers to output from a system, such as a video game that shows the user whether the user is performing well. For example, a character only proceeds to the next level when the client has pronounced the word correctly. In this case, the visual feedback is the character's movement from level A to level B.
- User Record Voice: The OSTS contains an automatic data collection tool that can record all activities while the client completes one of the exercises. Parents always need to give permission for this, which is why the feature is optional. When recording the voice of a patient consent is needed from the patient itself. In case of a minor, either the parents or legal guardians need to sign a consent form. If such a form is not signed before the therapy starts this feature cannot be used which would make online speech therapy without a live therapist being present during the session itself, rather difficult (Ben-Aharon, 2019). If permission is granted, the child and the parent can view the history of the audio recordings of the child. There is an option to listen to the recordings and practice again. The speech therapist can review the sessions' recordings and provide feedback during the next live session or by recording a video message and sending it to the child.
- Therapy exercises: The therapy exercises entail different speech exercises in a game format.

3.4.2. Speech therapist interface

The speech therapist interface feature is the display presented to the speech therapists when the OSTS is accessed by speech therapists. With this display, speech therapists can find all features that can assist them in doing their work. Below are brief explanations of some of the features available to speech therapists.

- Reports: The report generation module uses the service provided to the clients to create statistical analysis reports regarding the children's progress and performance during therapy sessions.
- Clinical Speech Therapy Repository: This contains valuable speech therapy resources and training materials that can be used by the speech therapists.
 - Ontologies: This feature provides information regarding the different categorisations of speech sound disorders. For example, an overview of the characteristics of the normal development of a client with stuttering designed using formal medical ontologies such as UML and sequence diagrams.
- Monitoring and activities: A module used to create skill plans for clients based on the assessment reports from the report generation module.
- Scheduling: The scheduling module contains a calendar that provides an overview of the scheduled sessions with different clients. Here speech therapists can also create, edit, or cancel sessions.
- User Data Management: This modules handles the information of clients. Below are some of the features available in this module.
 - Speech Language Anamnesis: A medical record specifying the diagnosis, general medical conditions and related disorders stated by doctors. This also contains a record of the cognitive development data, such as a gap in language development and the receptive language age as estimated by speech therapists.
 - Evaluations: The results of the clients during the therapy sessions are stored and presented.

- Personal data: Each client has their account details. This module provides the account information of these clients, such as username and personal information like age, educational level, and family names.
- Manager interface: The manager interface is optional and provides tools for HR, financial accounting, and planning. It is optional as there are already some existing systems for such activities, so it does not necessarily need to be included in the OSTs. The most important module of this interface is the report module which is used to create skill plans for clients based on the assessment reports from the report generation module.

3.4.3. Data processing

This module receives the data mostly audio files and pre-processes them for the data analysis module.

- Batch processing or real-time processing: For collecting audio samples during the sessions, real-time processing or batch processing can be used. During batch processing, audio data is collected and stored when a child is doing an exercise. Then the system processes the data periodically, for example, at the end of the day. However, if the system needs to provide the child with feedback during the exercise, it is more feasible to use real-time processing. For the expert system to analyse the data for errors, the recorded speech must be pre-processed immediately.
- Pre-processing: First, the data needs to be pre-processed, which can be done with the following two open-source tools (Franciscatto et al., 2021):
 - SOX (Sound eXchange v14.4.1) can be used for removing noise.
 - FFmpeg (v.3.2.12): this tool can be used for detecting sentences and measuring audio length.

3.4.4. Data analysis

An expert system is required for recommending therapy activities. Such a system is designed with the help of rules. The rules define the path the system should take according to the registered data. This module mostly has the following features explained below.

- Interaction management: When a child is taking a therapy exercise, the system receives feedback with the help of:
 - Automatic Speech Recognition: ASR can transform the acoustic microstructure of speech signal into its implicit phonetic macro-structure. In other words, a speech-to-text conversion where the output is a digital text corresponding with the recognised speech (Ghai and Singh, 2012). ASR, also called speech to text, entails converting spoken words into data that can be further processed, often in a textual format (Gaikwad et al., 2010). A typical ASR system consists of an acoustic analysis component and a language model (Yin et al., 2015). When people speak, they breathe vibrations in the air. An analogue-to-digital converter (ADC) transforms the sound waves into binary data that the machine can understand. The ADC filters the speech to remove unnecessary noise and normalises the sounds and speed of the speech to match the pre-recorded samples. Once the data is separated into different speech frames, it extracts the acoustic features representing all the information in this signal. The features of each frame are the input for the next component of the ASR system, which is the acoustic model (Padmanabhan and Johnson Premkumar, 2015). The acoustic model converts the audio signal into phonemes. A phoneme is a discrete, or a speech sound, and distinctive unit of a language that can differentiate between words (Kazanina et al., 2018). All vowel sounds have different frequency patterns, which can be pre-programmed into the computer to recognise when a spoken sound matches a specific vowel sound. Phonemes are minor elements of a language that make up its base (Kazanina et al., 2018).
 - Error Detection: Identification of errors using speech analysis algorithms by analysing produced vowels and consonants individually (Parnandi et al., 2013).
 - Classification: Identifies the audio input from the client as correct or incorrect (Franciscatto et al., 2021).
 - Feature extraction: Breaks the captured speech down into bits to be converted into a digital format (Belen et al., 2018).
- Face detection: Image detection of the face of the client through a webcam.
- Emotion screening: The system considers the client's emotion throughout the session, for example, by detecting the facial expressions with a face tracker or the system asking the child how he/she feels or rating the feelings of the child.
- Recommendation strategy: Has the main task of presenting the processed results and recommendations. In the study of Ochoa-Guaraca et al. (2016), this is done with an automated progress monitor, a client's profile analyser and a therapy template assemble

4. Reference architecture

The following paragraphs describe the selected viewpoints that were used for documenting the RA. The OSTs RA is named Talking Heroes. Consequently, Talking Heroes is portrayed based on these viewpoints. The features discussed in the views below are described in detail in Section 4.4 above.

4.1. Selection of views

An architectural framework organises and structures the proposed viewpoints (Tekinerdogan and Uzun, 2019). A widely used reference architecture framework is the “Views and Beyond” approach proposed by Clements et al. (2003). They distinguish between three different categories of viewpoints and further divide them into 17 subcategories. The RA should handle all features that are considered helpful from the previously discussed RAs. First, a context view is presented to define the scope of the system within the external environment (Tummers et al., 2021). Secondly, a decomposition view is provided to describe the composition of the key elements of the system. Unified Modelling Language (UML), a semi-formal graphical language is used for describing the system. OMG (2003) The UML description provides a conceptual model of the OSTs.

4.2. Context view

A context view is generally represented with a context diagram (see Fig. 3). The context view shows the relation of the system with entities outside the system’s scope. It provides information regarding what is included and what is not in the system and the overall purpose of the system (Tummers et al., 2021). The top-level context diagram scopes the entire system and provides a basis for the design of the additional architectures. The OSTs developer develops the OSTs together with the administrator who sets up the system. The child provides audio input to the system when completing exercises displayed on the user interface (UI) system. The audio input is then transferred to the Expert System in the Business Logic Layer, which has an Automatic Speech Recognition Module. Here, the audio file is also pre-processed to remove any background noise and ensure the audio file’s quality, (Franciscatto et al., 2021). After the expert system classifies the sound as wrong or correct, it sends feedback (audio, textual or visual format) to the child (client) during the exercise. The classification by the expert system is also periodically checked by the speech therapist. When the speech therapist concludes the classification was done correctly, it is transferred from the Expert System to the database where it is stored. When the child completes the session, a report is automatically generated by the Expert System, which contains information about the performance of the child and the progress during the therapy session. The speech therapist also manages the accounts of clients here, such as usernames and can access the personal data of clients. There is also a calendar feature where speech therapists can plan a live session with their client or schedule a specific exercise the client must do at that moment. Additionally, parents can see an overview of this in their calendar.

4.3. Decomposition view

Fig. 4 shows the decomposition view, which illustrates the decomposition of the system into different modules by embedding modules in the overall system module. The decomposition view is usually the foundation for a project organisation and documentation (Van Vliet et al., 2008). With this view, all possible but optional modules are presented. These eighteen (18) modules were confirmed by consulting a real speech therapy organisation (see appendix). This indicates that designers of an online speech therapy system should consider including all these modules (features). These modules are interconnected where data is exchanged or transferred between them. An important module is the voice recorder which is responsible for recording the sound (voice) of the end-user which will be sent to the Data processing and Data analysis modules. Error detection and feedback modules are also key to the feasibility of the OSTs. These modules detect errors in the speech of the end-user and provide important and reliable feedback to the end-user. In addition, security management is a vital module which ensures that personal and private data records of all users are protected from hackers or malicious users. The other modules are responsible for handling functionalities as suggested by their names.

4.4. Deriving AA from the RA

As Köksal and Tekinerdogan (2019) noted, to derive a concrete architecture for a particular context, different features of an OSTs should be selected. The approach used for deriving an OSTs application architecture is based on the product line engineering process approach used by Köksal and Tekinerdogan (2019), which consists of domain engineering and application development. Domain Engineering consists of three steps. The first step of the domain engineering activity of the proposed approach is the development of a family feature diagram for an OSTs that uses Automated Speech Recognition (ASR). A feature diagram is a tree model that shows common, alternative and optional features of a system (Tummers et al., 2021) Optional features can be selected but are not necessarily required, while alternative features require selecting at least one defining feature (Köksal and Tekinerdogan, 2019). The second step entails the development of an RA for an OSTs. Application Development consists of three steps. The first step entails developing an OSTs Application Feature Model. The application features are selected with the help of a case study. The second step is to develop the OSTs Application Architecture based on the RA created (Talking Heroes) during the domain engineering step. For the final step, the OSTs Application is implemented with the help of a simulation system.

The existing RAs discussed in the background section are too abstract to derive an application architecture from directly. Based on multiple existing studies, an OSTs family feature model was created.

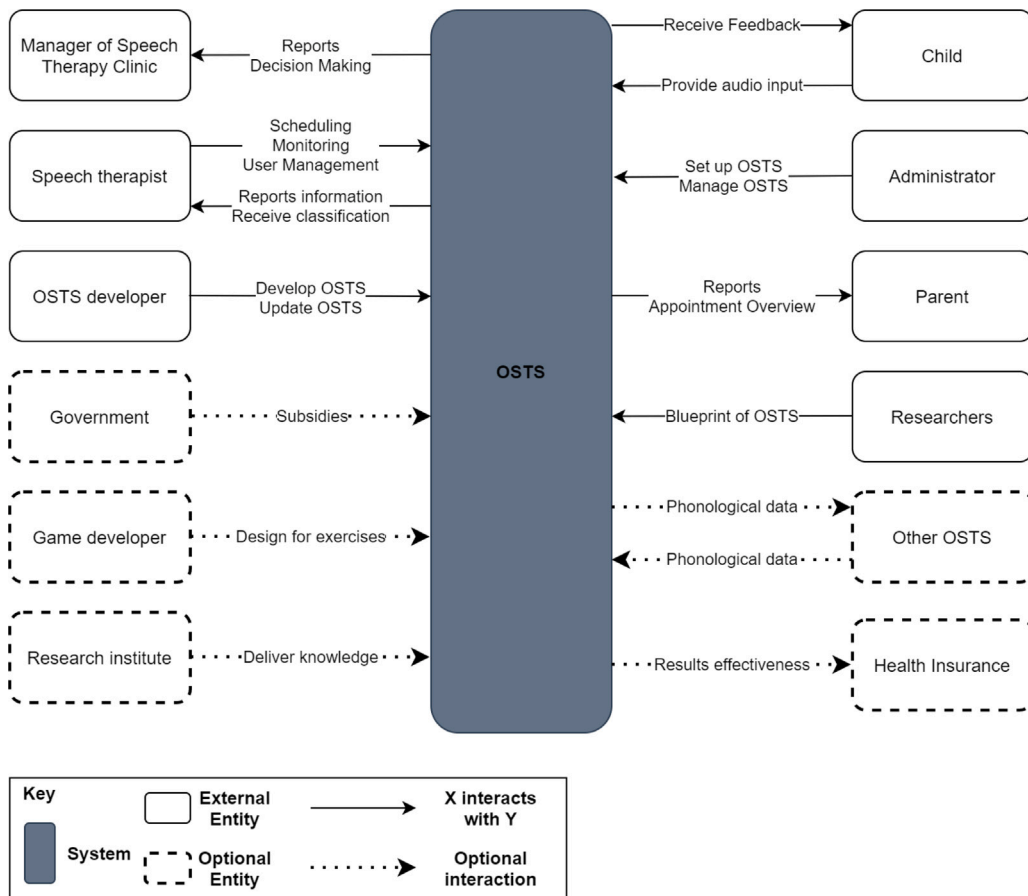


Fig. 3. Context view of the reference architecture.

5. Evaluation of the reference architecture: Case study approach

In this section, we evaluate the architecture designs, feature diagrams and business processes using a case study. The case study approach was based on the guidelines from [Runeson and Höst \(2008\)](#). This section follows the structure of their proposed steps: (1) case study design, (2) preparation for data collection, (3) data collection, (4) analysis of collected data, (5) reporting. Each step is further explained in the following sections. However, step 3 is discussed in [Appendix B](#) due to the amount of information.

5.1. Case study design

The first step of a case study is selecting a case, defining an objective, and making a plan. The case study research was applied to a prospective case. A retrospective case includes a system developed before and for which there is already an architecture with required design documents. The prospective case study includes the system that is planned to be developed ([Köksal and Tekinerdogan, 2019](#)). The objective of the prospective case is to evaluate the effectiveness of the designed RA and develop an AA for the company selected for this case study. The objective of the prospective case study is twofold:

1. Validation of our RA.
2. Conducting an industrial case study.

This was done with the help of the design activities presented in [Table 3](#). The case study was conducted in collaboration with an online speech therapy clinic based in the Netherlands with branches across the globe. The company is called TinyEYE and they provide speech therapy with the help of video communication software. Both the General manager and Chief Executive Officer (CEO) of this company participated in the study. The goal of the company is to provide their current services to a higher number of people without a reduction in quality. They are currently working on developing an feasible OSTs they could use to extend their services given the current COVID-19 pandemic.

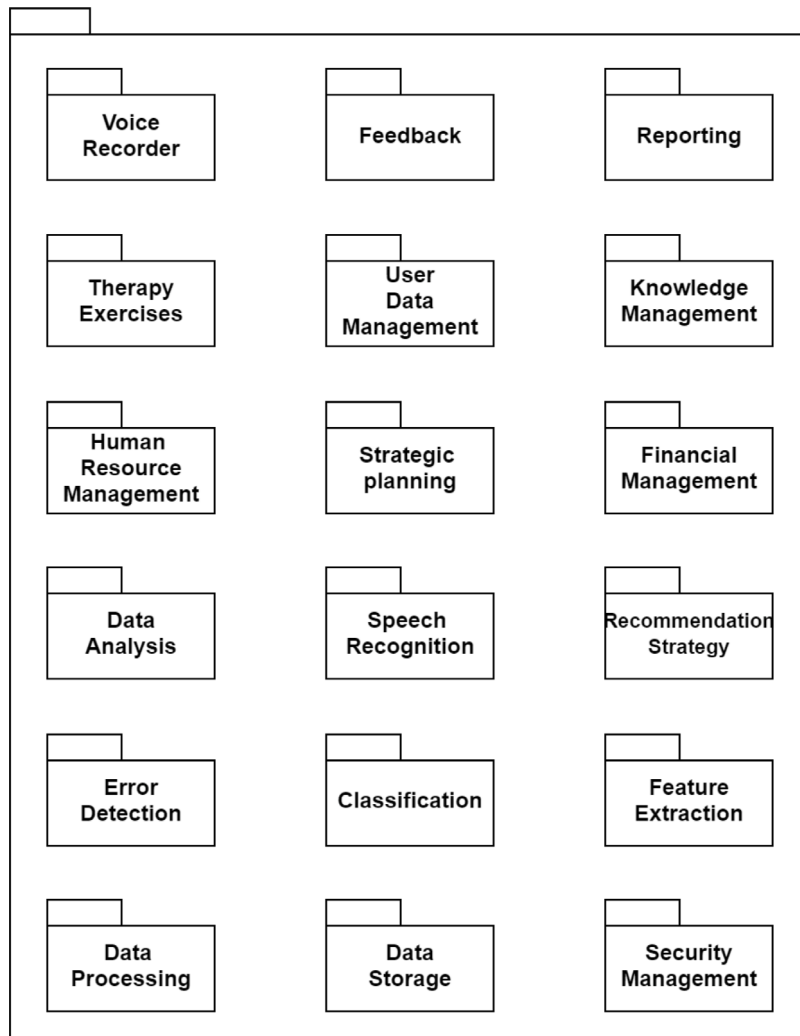


Fig. 4. The decomposition view of the reference architecture.

5.2. Preparation for data collection

This section describes the protocols and procedures for the data collection. A case study protocol explains the data collection procedures (Runeson and Höst, 2008). By conducting a semi-structured interview, we aimed to determine if the established RA was effective and useful and whether the RA was practical. We also created a documented questionnaire and asked TinyEYE to complete it so as to gather requirements for the AA. Both the questionnaire and the semi-structured interview questions can be found in [Appendix A](#). The selection, participation and execution of the interview can be found in [Appendix B](#).

5.3. Reporting

This section communicates the findings of the study and discusses the feasibility of the RA. The AA is derived and presented using the previously discussed online speech therapy clinic (TinyEYE) that provides speech therapy with the help of video communication software.

5.3.1. Feature diagram

To design the feature diagram, the CEO and General manager of the company were interviewed on some general questions to obtain a better idea of their understanding of an OSTS and how exactly it could help them. They describe an OSTS as an intelligent system that can detect and respond to the need for online speech therapy. The OSTS begins by identifying what is wrong with the client and then prescribing the correct exercises. The system must learn from the audio-visual feedback and adjust the exercise program accordingly. An OSTS can be considered as an extension of human intelligence. An OSTS is thus, like an assistant for the

Table 3
Case study design activities for Prospective case study.

Design activities	Description
Goal	Evaluate the RA for OSTs and derive an AA for a tailored OSTs.
Research Question	How effective is the designed RA for deriving AAs? (Section 5)
Data collection	Semi-structured interviews and architecture design.
Data analysis	Analyse the identified shortcomings and differences between AA and RA.

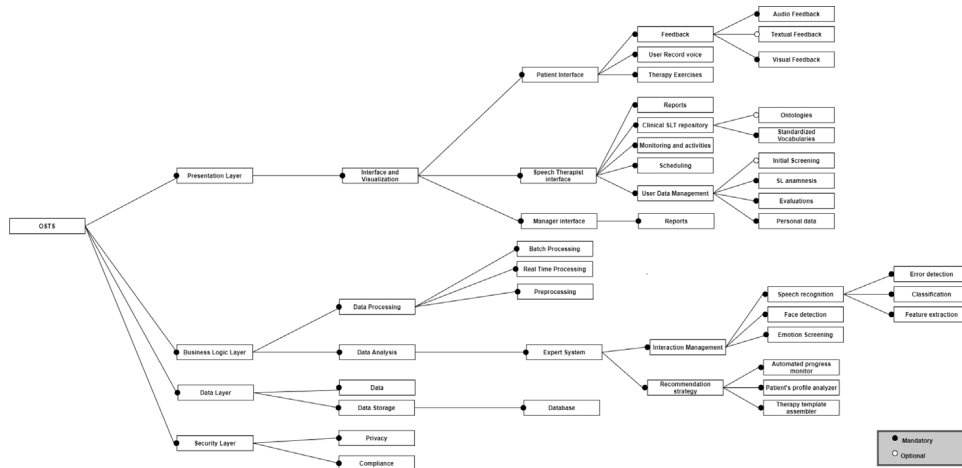


Fig. 5. Application feature diagram TinyEYE.

speech therapist since the intake and diagnosis will initially always be made by a live speech therapist. The speech therapist will always compose the therapy plan. The intelligent system provides all the exercises to the clients and evaluates the exercises, however the outcome is assessed by the speech therapist. In a later stage, the system might even be able to make a diagnosis, but for now, it is still the speech therapist who performs this. Based on the results from the questionnaire (see [Appendix C](#)), a feature diagram was made (see [Fig. 5](#)).

5.3.2. Context diagram

[Fig. 6](#) presents the context diagram for the case study. In comparison with [Fig. 3](#), this diagram has more external entities. With the help of the semi-structured interview, several new external entities came up. During the interview, TinyEYE mentioned that a parent feature is not comprehensive for all their cases. As an example, there are children without parents or clients where the teacher at school acts as an internal supervisor of the child about speech therapy. When there is an adult client, there may also be a caretaker who, for example, cares for a client with Parkinson's and plans the appointments for him or her. Within TinyEYE there are two different types of management. First, the general management, but secondly, the speech therapists also have their manager, the speech therapist's manager. When starting a new therapy plan with a client, this is sent to a medical doctor, who must accept it for the treatment to be covered by the health insurance. It would be convenient if the medical history regarding the client's speech could be sent straight from the medical doctor (this could be the hospital but also a General Practitioner (GP)). This ensures that TinyEYE already has some basic and necessary information when starting with a new client. However, this does not mean they would not make their diagnosis. The research specialists and educational institutions can provide domain knowledge for the OSTs to be financed. Some other changes were also made. The entity "child" was changed to client, as TinyEYE also offers therapy to adults and not only to children. Government and health insurance became mandatory instead of optional as both parties are necessary for financing and policies regarding the health care.

5.3.3. Decomposition view

TinyEYE would like to see the following ten functions in the system as well:

1. Video recorder to enable face detection and emotion screening.
2. Ongoing learning system that will automatically learn from the user input and provide customised feedback. A continuous learning system is possible with the integration of Artificial Intelligence (AI) and Machine Learning tools.
3. Avatar game character live practising during the session
4. Customisation of the virtual therapist avatar would allow the client to personalise the application by changing the avatar to their favourite one.
5. Digital human module that can produce avatars with the possibility of imitating human body language and providing non-verbal responses. With the advent and prevalence of AI tools, the design and development of this module is possible.

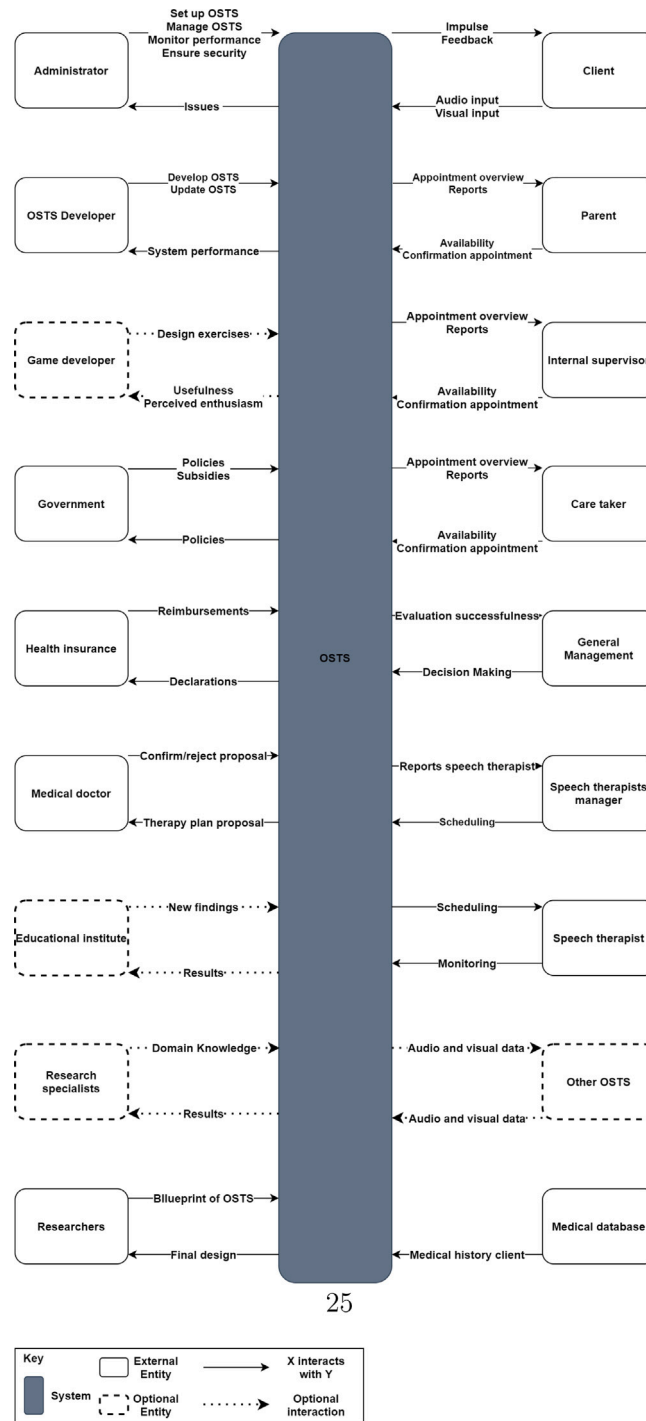


Fig. 6. Context diagram TinyEYE.

6. Saving a therapy session so the client can practice individually. Based on the session with the avatar, the child can now practice individually without the need for the constant presence of an avatar.
7. Registration of how often the client practices individually.
8. Progress evaluation of each client with reliable reports presented with graphs and tables.
9. Feedback and reward system (e.g., digital sticker)

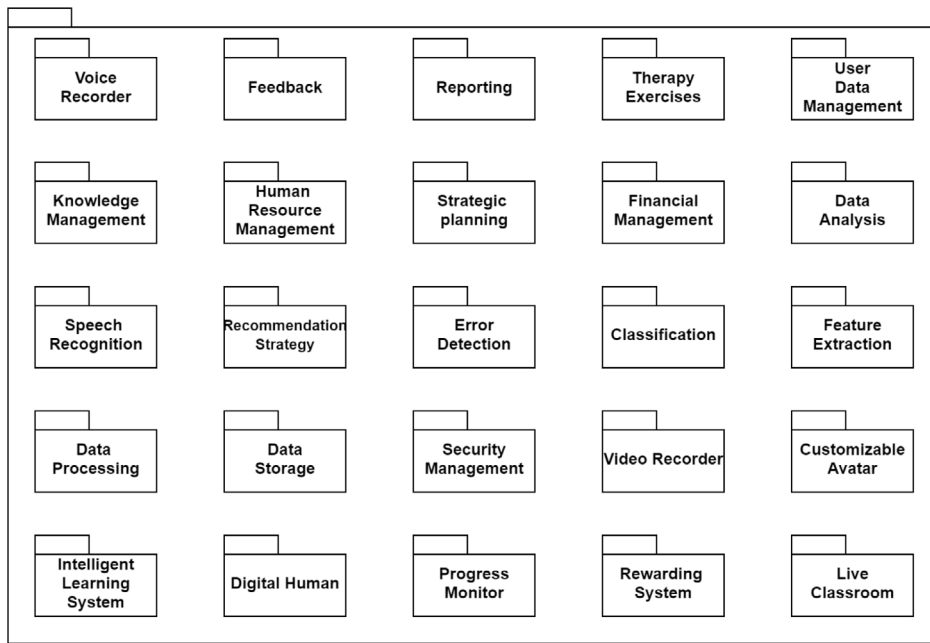


Fig. 7. Decomposition view of TinyEYE.

10. Attention measuring of the client during the sessions with the help of AI so that when attention decreases, a short video is shown, for example.

The above described modules resulted in the creation of the decomposition view presented in 7.

5.4. Overall results and findings

In this section, we discuss the effectiveness and practicality of the designed and developed RA. We answer the main research question “*What is a proper reference architecture for OSTs?*” The main challenge we faced was how to obtain a benchmark RA to be used for comparison. Unfortunately, there were no existing RA designed for the speech therapy domain according to the literature and the industry partners also did not have any architecture document available during our interview. Thus, we ensured the industry respondents (Manager, CEO and external software architect) received our designed diagrams days ahead of the interviews. Although the manager and CEO of the clinic were not familiar with the feature diagram and the architecture view modelling approach, explaining this did not require much effort; however, there were some unclarities. When asked if the context view and decomposition view were understandable, the interviewees gave feedback that it would have been easier to understand if, in the beginning, the link between the feature diagram and the context diagram and decomposition view had been explained. Consequently, in the future, it would be beneficial to first discuss the feature diagram as the context diagram does not display the complexity of the OSTs. The RA brought up many questions at the beginning of the interview regarding any missing features. It was also unclear why there were no interactions between the entities themselves, which motivated us to develop a use case diagram (see Fig. 8).

Furthermore, the initially designed context diagram appeared to be incomplete as some modules had to be added. They also remarked that a layered view was missing. Before the interview, we did not think it would be necessary to discuss the layered view from Fig. 1, but it would have been better to do so. Notwithstanding all the initial challenges, the interviewees appreciated the designed RA and derived AA. From the interview questions “*Is the designed architecture similar to/ better than your previous architecture document used for your current speech therapy software system?*” and “*Will you use this architecture?*”, the respondents gave a high score of 8 out of 10 for the effectiveness of our designed approach and 9.5 out of 10 for the practicality of the designed architecture. The respondents acknowledge that the designed architecture was significantly better than the architecture used for their old software system. In addition, the high score of 9.5 out of 10 for the practicality of the designed architecture indicates their appreciation and acceptance of the designed RA. The concerns of the manager and CEO were rightly incorporated into the designed RA, AA and views as presented in Figs. 1–7 above. They deemed the designed architecture as more practical and would use this product for the new software system. In summary, the answer to the question “*What is a proper reference architecture for OSTs?*” is provided below. *After conducting a case study and several iterations with the industry practitioners, the designed reference architecture can be considered as a solid and feasible architecture for an online speech therapy system.*

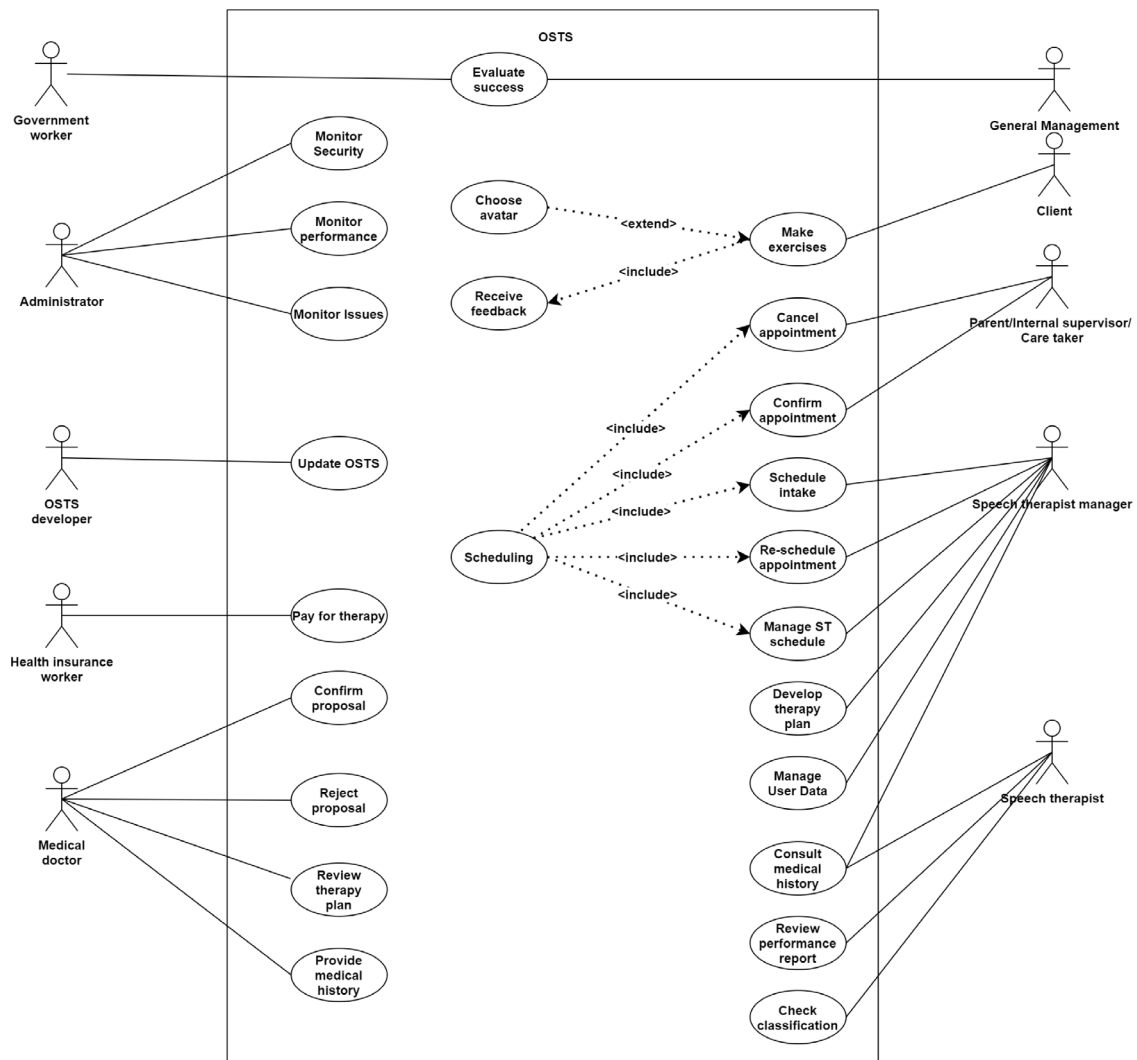


Fig. 8. Use case diagram TinyEYE.

6. Discussion

This study designed and presented a novel RA for online speech therapy systems based on inputs from the prior literature and refined after conducting a case study with a speech clinic based in the Netherlands. Well-established architectural models and approaches were followed in designing the RA and AA to answer these two main sub questions as follows:

1. Is the designed architecture similar to/ better than your previous architecture document used for your current speech therapy software system?

2. Will you use this architecture?

The initial goal for the case study was to apply the RA for both a prospective case and a retrospective case. The goal of the retrospective study was to compare the architecture of their developed system with an AA for this system made by us. However, during the design of the AA, it became clear that without official required documents or official architecture design documents, this was neither feasible nor useful. As the case study interview was not recorded, there was a risk of inaccuracies. To avoid this, the transcripts of the meeting were sent for a check to the company. A thorough analyses of the answers provided to the interview questions indicated that the designed RA was more effective and better than their previously designed architecture. Regarding the second question, the respondents were satisfied with the designed approach and stated that they will definitely use the designed architecture for their new software system. By evaluating our designed RA with industry practitioners, we observed that the designed RA is feasible and reliable for a new speech therapy software system. Based on the analyses, we answer our main RQ “What is a proper reference architecture for OSTS?” We designed a robust and usable RA after several iterations with the industry practitioners through the case study and the designed RA is a more feasible and reliable RA for the OSTS domain. This study complements some

prior studies (Franciscatto et al., 2021; Ochoa-Guaraca et al., 2016; Redrovan-Reyes et al., 2019) that designed system architectures based on abstract modelling techniques. Our study incorporates the feedback of some OSTS industry practitioners to develop a more applicable and formal reference architecture. We thus recommend future studies to always evaluate their theoretical frameworks with industry practitioners.

Similar to previous studies (Franciscatto et al., 2021; Ochoa-Guaraca et al., 2016; Redrovan-Reyes et al., 2019), Talking Heroes is based on a layered RA shown in Fig. 1. Since we conducted a domain analysis based on previous studies, the presentation layer comprises of several visualisation features for clients and speech therapist. Talking Heroes, however provides more formal details on the interaction between the layers different from the abstract architectures provided by these previous studies. Additionally, the proposed RA provides the opportunity to derive an Application Architecture (AA) and a family feature model was created.

To generalise the results obtained at the case study, it would have been necessary to conduct more case studies. The decision not to do so was due to time constraints and the limited presence of online speech therapy clinics on the market. Another limitation of the research is that the final architectural evaluation is somewhat limited. For future work, the OSTS can be implemented by designing a simulation system. A proof-of-concept is necessary to determine if the established AA can be implemented to achieve the requirements. Additionally, for further implementation, a proof-of-technology would be needed to identify in more detail which selected technology to use. Some technological options are provided for the OSTS family feature diagram but not for the final AA itself. The current AA would also benefit from a further analysis regarding its security, reliability and re-usability to ensure the quality of the AA. This could be, for example, by proposing more scenarios during a more in-depth case study interview to the CEO and Manager of TinyEYE.

7. Conclusion

Online speech therapy systems (OSTSs) can be a feasible solution for the current shortage of speech therapists. Developing these systems are require formal and well-designed reference and application architectures which are designed by system analysts. Existing system architectures designed and developed for OSTSs are informal and do not follow any software architecture documentation guidelines. This study has provided a reference architecture and an AA for an OSTS. A domain-driven approach was used to support the design of the application architecture, and a family feature diagram was made. The case study approach showed that the reference architecture can be easily used to develop an AA for OSTS. The adoption of the reference architecture helped also to increase the development time and ensure that the proper modules are selected for the OSTS. The proper design of the architecture using formal architecture views helped to support the communication regarding the design decisions. In our future work we will apply the reference architecture for the development of other OSTSs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

Special thanks to Elisa Gielen (Manager), Paul Jenkins (CEO) and Mila Rozema (Speech Therapist) of TinyEYE, the Speech Therapy Clinic based in the Netherlands for offering to use their CLINIC as a case study.

Appendix A. Interview document for TinyEYE

A.1. Informed consent form

The interviews you are being requested to participate in are part of a research on the design of application architecture for an Online Speech Therapy System (OSTS). The purpose of the interviews is to obtain a detailed view on the requirements of the to be designed OSTS. The case study approach first aims to access how effective the designed general reference architecture for OSTS is for deriving an application architecture for an OSTS. The meeting will approximately take around 1 to 1.5 h, depending on your availability.

- Your participation in the study is entirely voluntary. Thus, you are in no way obligated to participate in it. You may ask questions about the study both before committing to participate in it and at any time throughout the study.
- You may also refuse to answer any question at any time, and you may choose to end the interview or withdraw from the study at any time.
- A written transcript of the interview will be made to apply content analysis.
- The information we will be asking you to share WILL be published in a research report. However, the identifying data about the individual employees, including names or other identifying factors, will be anonymised for publication purposes.
- You will also be asked to review and approve any written materials before they are published.

A.2. Interview 1

The aim of this interview is to get a first impression regarding the practicality and usefulness of the designed general Reference Architecture. We have developed a context diagram and a decomposition view.

A.3. Context diagram

The context view shows the relation of the system with entities outside the system's scope. To a reader, it provides information regarding what is in and what is out of the system and the overall purpose of the system.

- Do you think there are any stakeholders missing?
- Apart from TinyEYE, do you think external parties could use the system?
- Apart from TinyEYE, do you think external parties could benefit from the system?
- Can you check the relations? Are these correct?

A.4. Decomposition view

The decomposition view defines the breakdown of different modules in the system. This view can help to distract the user requirements as it gives a list of things you are supposed to ask when designing a software architecture.

- Which functions should the system have?
- Do you think some of the above functions are unnecessary?
- What should the system look like?
- What are the mandatory features? What are optional?

A.5. Concluding questions

- Are the Context View and Decomposition view understandable?
- How do you think the current architecture can be improved/become more understandable?
- Is the designed architecture similar to/ better than your previous architecture document used for your current speech therapy software system?

A.6. Questionnaire 1

With the help of this questionnaire, we are trying to gain a better insight regarding the requirements of the to be developed Application Architecture. This questionnaire consists of two parts. The first parts are some general open questions to get a better idea of the way TinyEYE looks at the concepts of Online Speech Therapy Systems (OSTSs) itself.

A.6.1. Part 1: General questions

Please fill in your answer in the textbox

1. How would you describe an OSTS?
2. What would you consider most important for an OSTS?
3. What are the main characteristics of your clients regarding their:
 - a. ..age?
 - b. ..educational background?
 - c. ..parental support?
 - d. ..speech disorder?

A.6.2. Part 2: Application Feature modules questionnaire

Please select for the [Table A.4](#) shows the following Features whether you would like to see it back in the system or not. If you are not sure please select the box Maybe and explain in the comments why you are not sure about this feature.

Now for each feature that you selected please answer the following questions:

- When will this feature be used?
- Who will use this feature?
- Who will receive the outputs of the feature? Who will learn about the results of someone using this feature?
- Are there any features which you have missed in this list?

Table A.4
Application Feature modules questionnaire.

Feature	YES	Maybe	NO	Comments
Client Interface	☐	☐	☐	
Feedback	☐	☐	☐	
Audio feedback	☐	☐	☐	
Textual feedback	☐	☐	☐	
Visual feedback	☐	☐	☐	
User record voice	☐	☐	☐	
Therapy exercises	☐	☐	☐	
Speech Therapist Interface	☐	☐	☐	
Reports	☐	☐	☐	
Clinical ST Repository	☐	☐	☐	
Ontologies	☐	☐	☐	
Standardised Vocabularies	☐	☐	☐	
Monitoring & activities	☐	☐	☐	
Scheduling	☐	☐	☐	
User Data Management	☐	☐	☐	
SL anamnesis	☐	☐	☐	
Evaluations	☐	☐	☐	
Personal data	☐	☐	☐	
Manager interface	☐	☐	☐	
Reports	☐	☐	☐	
Human Resource Management	☐	☐	☐	
Financial management and accounting	☐	☐	☐	
Strategic planning	☐	☐	☐	
Decision making	☐	☐	☐	
Batch Processing	☐	☐	☐	
Real time processing	☐	☐	☐	
Pre-processing	☐	☐	☐	
Expert System	☐	☐	☐	
Interaction management	☐	☐	☐	
Speech recognition	☐	☐	☐	
Error detection	☐	☐	☐	
Classification	☐	☐	☐	
Feature extraction	☐	☐	☐	
Face detection	☐	☐	☐	
Emotion screening	☐	☐	☐	
Recommendation strategy	☐	☐	☐	
Automated progress monitor	☐	☐	☐	
Client's profile analyser	☐	☐	☐	
Therapy template assembler	☐	☐	☐	
Database	☐	☐	☐	
Privacy	☐	☐	☐	
Compliance	☐	☐	☐	

Appendix B. Selection and participation

The case study was conducted in collaboration with an online speech therapy clinic named TinyEYE. TinyEYE is an online speech therapy clinic that provides speech therapy with the help of video communication software. Both General manager of TinyEYE and CEO of TinyEYE, were keen to participate. TinyEYE wants to provide their current services to a higher number of people without the quality reduction. They are currently working on developing an OSTs that they could use to extend the services that they are now providing.

Table C.5

Application Feature modules questionnaire.

Feature	YES	Maybe	NO	Comments
Patient Interface		☐	☐	
Feedback		☐	☐	
Audio feedback		☐	☐	
Textual feedback	☐		☐	Alleen voor volwassenen en oudere kinderen.
Visual feedback		☐	☐	
User record voice		☐	☐	
Therapy exercises		☐	☐	
Speech Therapist Interface		☐	☐	
Reports		☐	☐	
Clinical ST Repository		☐	☐	
Ontologies	☐		☐	Hebben we nu ook niet.
Standardised Vocabularies		☐	☐	
Monitoring & activities		☐	☐	
Scheduling	☐		☐	Geïmporteerd, maar wel zichtbaar. Input komt van ander system. Niet efficient om twee keer te doen.
User Data Management		☐	☐	
SL anamnesis		☐	☐	Intakemoment, problematiek
Evaluations		☐	☐	Evaluation is output.
Personal data		☐	☐	Hoeft niet de volledige data te zijn, input uit een ander systeem.
Manager interface		☐	☐	
Reports		☐	☐	Informatie over het gebruik van het systeem en hoe efficient/effectief is het.
Human Resource Management	☐	☐		
Financial management and accounting	☐	☐		
Strategic planning	☐	☐		Kan de manager zelf doen op basis van reports.
Decision making	☐	☐		
Batch Processing		☐	☐	
Real time processing		☐	☐	
Pre-processing		☐	☐	
Expert System		☐	☐	
Interaction management		☐	☐	
Speech recognition		☐	☐	
Error detection		☐	☐	
Classification		☐	☐	
Feature extraction		☐	☐	
Face detection		☐	☐	
Emotion screening		☐	☐	
Recommendation strategy		☐	☐	
Automated progress monitor		☐	☐	
Patient's profile analyser		☐	☐	
Therapy template assembler		☐	☐	Input kan ook weer door human gedaan worden.
Database			☐	
Privacy		☐	☐	
Compliance		☐	☐	

The execution of the interview took place on the 19th of May in the morning. The meeting was held with one of the researchers, CEO, and General Manager using Microsoft Teams. Before the meeting, the document with the interview questions and questionnaire was sent beforehand to TinyEYE. Also, we went through the Informed Consent Form together, and the interview's objective was explained.

The interview was not recorded due to privacy reasons. However, notes were made while asking the questions by the researcher in a Word Document, which was also displayed to the General Manager and the CEO via screen sharing so they could see and verify

what was been written. The whole meeting took 2 h and 45 min, including a 10-minute break. Before the break, questions were asked about the context diagram and the decomposition view. After the break, some general open questions were asked to understand better what TinyEYE expects from an OSTs. Then, they were asked to fill in a questionnaire regarding the feature modules of the OSTs. Answering the questionnaire could have been done without one of the researchers being present during the meeting, but this was deliberately done to ask questions when a feature was unclear.

Appendix C. Results of the application feature modules questionnaire

See Table C.5.

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